

HARD DRIVES ARE DEAD – Or Are They?



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Sometimes it is the applications that drive a technology. We can look at 'Big Data' as one of the reasons why the spotlight is back on solid-state storage technology. Solid-state storage is not new. This silicon based electronic method of data storage has been around for a long time now. As 'memory' to speed up transactions between processors and hard disk drives, and then as 'storage' in smaller devices like mobile phones, solid-state storage has proved to be useful. It has also been used for mission-critical applications like defence where speed is imperative.

However, with a splurge in Big Data brought about by the Internet of Things (IoT) devices and enterprise applications, the need for faster storage is felt more than ever—giving rise to the development of mega-sized solid-state drives (SSDs) and increased research on how to make larger SSDs at a lower cost. While we might have thought it impossible some years back, online marketplaces show that it is now possible to buy a two-terabyte (TB) peripheral component interconnect express (PCIe) based SSD for less than US\$ 3000 and a 1TB one for US\$ 500-1000. So we might

not be entirely wrong in hoping for a future of data centres populated entirely with SSDs!

Terabyte-scale storage sees uptrend (literally)

The industry has been steadily increasing the capacity of flash memory by reducing space wastage, but still there is only so much you can store on a piece of silicon! However, even at a stage when it is difficult to reduce the cells' pitch size further, the capacity of SSDs continues to go up, thanks to technologies that enable cells to be stacked one above the other. In 2015, all the industry majors ranging from Intel Corporation and Micron Technology, SanDisk and Toshiba, to Samsung, unveiled prototypes that involved stacking of flash cells.

Basically, one of the ways in which flash memory has evolved in the past deals with the number of bits stored per cell. The earliest were single-level cells (SLCs) that stored one bit per cell, followed by multi-level cells (MLCs) that stored two bits per cell, and then triple-level cells (TLCs) that can store three bits per cell. Although storing more bits in a cell made flash memory cheaper and capable of storing more data, it was not very reliable. Overall, SLCs were found to be faster, more durable, reliable and power-efficient for large enterprise-scale applications. Last year, Intel and Micron revealed a new three-dimensional NAND (3D NAND) technology that uses MLC and TLC technology to bring down the cost of storage, while also promising enough reliability and power-saving for large applications.

Here, flash cells are stacked vertically in 32 layers to achieve 256-gigabit (32GB) MLC and 384-gigabit (48GB) TLC dies that fit within a standard package. According to the press release, this enables three times the capacity of existing 3D technology—up to 48GB of NAND per die—enabling three-fourths of a terabyte to fit in a single finger-tip-size package. So a small gum-stick-size SSD can hold more than 3.5TB of storage and a standard 6.35cm (2.5-inch) SSD can

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3D XPoint™ Technology is non-volatile, which means your data doesn't go away when your power goes away—making it a great choice for storage.

Stackable
3D XPoint™ Technology can be stacked to further boost density.

Non-Volatile
3D XPoint™ Technology is non-volatile, which means your data doesn't go away when your power goes away—making it a great choice for storage.

Selector
Whereas DRAM requires a transistor to write memory, 3D XPoint™ Technology enables the memory cell to be written to or read without requiring a transistor.

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3D XPoint™ Technology is not significantly impacted by the number of write cycles it can endure, making it more durable.

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For the first time, there is a fast, non-volatile and non-wear-out memory technology that can serve as system memory and storage.

~8x to 10x Greater Density than DRAM*
3D XPoint™ Technology's simple, scalable, transistor-less design packs more memory into less space while it's critical to meeting cost.

	1GB	16GB	32GB
3D XPoint™ Technology	1000	1000	1000
DRAM	1000	1000	1000

3D XPoint™ Technology Processor DRAM 3D XPoint™ Technology

(Courtesy: Intel)